



Insight Remote Support Security White Paper

Version 7.0.9

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Related Documents

- Insight Remote Support 7.0.9 Release Notes
- Quick Installation Guide
- Installation and Configuration Guide
- Managed Devices Configuration Guide
- HP Insight Online Direct Connect Architecture and Security Model: HP ProLiant Gen8 Servers and BladeSystem c-Class Enclosures

This document describes the security aspects of the HP Insight Remote Support solution and its components. It describes the security features and capabilities of the solution.

Overview

Today's IT department plays a central role in meeting business objectives. Leveraging your IT infrastructure investments and improving overall system availability and utilization are crucial in today's business environment. HP Insight Remote Support and Insight Online simplify the management of highly diverse IT environments by providing automated failure detection and reporting as well as advanced analytics for your IT environment. HP Insight Remote Support can automatically detect and analyze problems in your IT environment. If a repair is necessary, Insight Remote Support will automatically log a support case and dispatch it to HP Support for resolution.

Today, many security-sensitive transactions—such as e-commerce, stock trades, and online banking—are executed securely over the Internet using the same standard security technology utilized by HP through Insight Remote Support.

HP understands and shares your company's security and privacy concerns and has leveraged its experience as a technology leader to create a secure remote support solution.

Specifically, HP provides a multilevel, layered security structure through encryption, authentication, standard security protocols, and industry best practices integrated at the physical, network, application, and operational levels. Interactions between HP and your enterprise network are restricted and tightly controlled through a single, secure access point.

HP's remote monitoring and support capabilities, along with any support information collected, are used only to provide you with world-class HP support. To assure that your data is safe with HP, the Insight Remote Support solution has received the TRUSTed Web Privacy-Cloud certification.

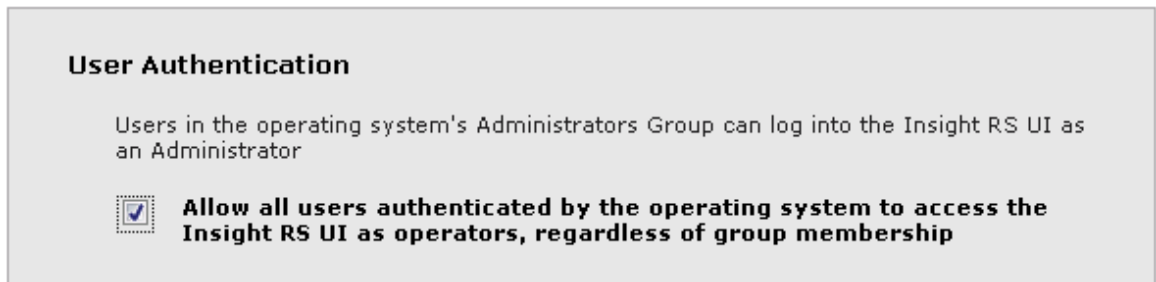


Insight Remote Support - Communications

There are several communication methods used in Insight Remote Support. These include: Device Discovery, Event Management, Data Collection, Data sent to HP, Data Management at HP, and accessing data using Insight Online.

Insight Remote Support User Interface

The Insight RS Console allows a system administrator to view configuration details about devices in their enterprise. User access to the Insight RS Console is controlled by the Windows account settings. Users in the Windows Administrator group will have full access in the Insight RS User Interface. Non-Administrator (operator-level) user access is disabled by default. Operator-level access to the Insight RS Console can be granted by checking the appropriate box in the Administrator Settings → Settings tab:



Operator-level users cannot perform the following administrative actions:

- Change the company name
- Change the web proxy used by the Hosting Device
- Change the 'opt-in' values
- Change the HP Insight Online/HP Passport settings
- Update Remote Support versions
- Trigger or configure discovery enterprise wide
- Enable or disable user authentication

Password management for the Insight RS Console is managed at the Operating System level. Local access passwords are not stored within the application. To access the Insight RS Console, open a browser window (see Release Notes for a full list of supported browsers), and browse to the URL: https://<hosting_device_IP>/Full_Domain_Name>:7906 and enter your user credentials in the Username: and Password: boxes.

HP SIM Adapter

If HP Systems Insight Manager (SIM) is installed on the Hosting Device, HP Insight Remote Support and HP SIM can share information via the HP SIM Adapter. If HP SIM is installed before Insight Remote Support, the HP SIM Adapter will automatically be installed with Insight Remote Support. If HP SIM is installed after Insight Remote Support, the HP SIM Adapter must be manually installed. Visit the [HP SIM Information Library](#) for more information on HP SIM.

Note: If HP SIM and the HP SIM Adapter are installed on the Hosting Device, it is important that you discover devices using HP SIM to ensure that devices are synchronized between HP SIM and Insight Remote Support.

Email Adapter

Insight Remote Support can notify the (default and backup) device contacts via email when certain events occur. Email notification can be configured in the Integration Adapters tab in Administrative Settings menu of the Insight RS Console. Device contacts can be notified for any or all of the following events:

- A support case has been opened at HP for a specific event
- A support case has been closed at HP for a specific event
- A configuration collection has been sent to HP
- The Insight Remote Support application has failed
- An Insight Remote Support software update is available
- A monitored device maintenance agreement is about to expire
- A new device has been discovered
- Hosting Device capacity threshold has been reached

Insight Remote Support Device Discovery

Device Discovery is used to scan your network, or a portion of your network, and search for potential devices to be included in Remote Device Monitoring (RDM) and the Remote Device Collections (RDC). Device Discovery uses standard network management protocols (like Simple Network Management Protocol – SNMP and Web-based Enterprise Management – WBEM) to identify devices connected to your network (see: Chapter 3 of the Installation and Configuration Guide). For each device discovered on your network, the discover engine will attempt to communicate with it using all available services (listed in table 1). The discovery will identify all possible communication methods and allow the administrator to select the best one for the target device. Network management protocol passwords are encrypted using Advanced Encryption Standard 128 (AES-128) and stored in the Insight Remote Support database on the Hosting Device. These protocols must be configured in the Discovery → Credentials tab prior to discovery in order to properly identify devices on your network (assuming you are not using HP SIM discovery to identify devices on your network). The discovery engine can add devices the following ways.

Scan using an IP address, a list of IP addresses or range of IP addresses

This is the most restrictive method, allowing the administrator to target specific devices or a specific subnet range for device discovery.

Scan using a Windows Domain Group

This allows the Hosting Device administrator to discover all devices in a Windows Domain. This option requires the Windows Domain administrator username and password to be configured in the discovery engine on the hosting device.

Scan a LAN Subnet or all LAN Subnets

This allows the Windows Administrator to automatically discover all devices contained in a LAN Subnet. The Subnet range is defined by the LAN IP/Network Mask. This can be the entire network (or networks) or any subnetwork range within your enterprise with IP Routing enabled from the Hosting Device.

Device Discovery will scan your network for possible monitored devices using all available management protocols. These include:

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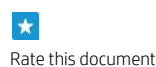


Table 1: Device Discovery Services

Service	Protocol/Port	Source	Destination
DCOM*	TCP/135	Hosting Device	Monitored Device
ELMC	TCP/7920	Hosting Device	Monitored Device
HTTP*	TCP/80	Hosting Device	Monitored Device
HTTPS	TCP/443	Hosting Device	Monitored Device
P4000 CLI	TCP/5989	Hosting Device	Monitored Device
P6000 CV	TCP/2372	Hosting Device	Monitored Device
RIBCL	TCP/443	Hosting Device	Monitored iLO Device
SNMPv1*	UDP/161	Hosting Device	Monitored Device
SNMPv2*	UDP/161	Hosting Device	Monitored Device
SSH	TCP/22	Hosting Device	Monitored Device
Telnet*	TCP/23	Hosting Device	Monitored Network Device
WBEM	TCP/5989	Hosting Device	Monitored Device
WS-MAN	TCP/443	Hosting Device	Managed Superdome OA
WMI#	TCP/135	Hosting Device	Monitored Windows Server

*DCOM, HTTP, PING, SNMPv1, SNMPv2 and Telnet are unencrypted protocols

#WMI is a DCOM service to configure your firewall to support DCOM services see:

<http://support.microsoft.com/kb/832017>. To restrict WMI to a specific port see: <http://msdn.microsoft.com/en-us/library/windows/desktop/bb219447%28v=vs.85%29.aspx>

Communication Services

The following services are used by HP Insight Remote Support for one or more of the following tasks: Remote Device Monitoring (Device Discovery and Event Notification); Remote Data Collection; Remote Device Access.

DCOM

The Distributed Component Object Model (DCOM) is a Windows protocol that enables software components to communicate directly over a network. Previously named "Network OLE", DCOM is designed for use across multiple network transports, including Internet protocols such as HTTP and WMI (Windows Management Instrumentation).

DCOM allows processes to be efficiently distributed to multiple computers so that the client and server components of an application can be placed in optimal locations on the network. Processing occurs transparently to the user because DCOM handles this function. Thus, the user can access and share information without needing to know where the application components are located. If the client and server components of an application are located on the same computer, DCOM can be used to transfer information between processes.

ELMC

The Event Log Monitoring Collector (ELMC) is a proprietary management service included with Insight Remote Support. ELMC is platform-specific and provides error condition detection on the monitored endpoint system on which it is installed. It communicates these events to Insight RS on the Hosting Device, which can be running either on the same system as the ELMC system or another system on the same TCP/IP network. Different ELMC packages exist for the same ELMC version, depending on the operating system and hardware platform.

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ESP

Encapsulating Security Payload (ESP), or IP protocol 50, is a protocol header inserted into an IP datagram to provide data encryption and authentication. Remote Device Access uses ESP in tunnel mode to establish VPN connectivity. ESP is described in [RFC 4303](#).

HTTP

The Hypertext Transfer Protocol (HTTP) is an application-layer protocol used for exchanging data. HTTP is described in [RFC 2616](#). Its most popular usage is for transferring text, graphic images, sound, video, and other multimedia files to Web browsers. HTTP capabilities are also general enough for non-web applications. HTTP communications are unencrypted. HTTP typically uses Transmission Control Protocol (TCP) port 80. HTTP is used by Insight Remote Support to discover monitored devices and communicate with older network devices that do not support encrypted communications.

HTTPS

HTTPS is HTTP with Secure Sockets Layer (SSL) or Transport Layer Security (TLS) encryption for security. All communications between the Hosting Device and the HP Remote Support Data Center are carried out over HTTPS. HTTPS is also used for the marshalling and transfer of collected device data between the Hosting Device and the monitored systems. HTTPS typically uses TCP port 443, but other services, like Remote Insight Board Command Language (RIBCL) and Web-Based Enterprise Management (WBEM), may specify a different port number for HTTPS communications. HTTPS is described in [RFC 2818](#).

IPSec

IP Security, or IPSec, is a suite of protocols for securing IP communications. IPSec operates in two modes. In transport mode it can be configured to provide end-to-end security of all communications between two systems. In tunnel mode, IPSec can be used to provide Virtual Private Network (VPN) connectivity over insecure networks. A typical IPSec deployment uses two protocols: Internet Security Association and Key Management Protocol (ISAKMP) and either Encapsulating Security Payload (ESP) or Authentication Header (AH), both of which are IP protocols. AH is seldom used as it does not provide encryption. IPSec is described in [RFC 4301](#).

IKEv2

Internet Key Exchange version 2 performs mutual authentication between two parties and establishes an IKE security association (SA) that includes shared secret information that can be used to efficiently establish SAs for Encapsulating Security Payload (ESP) [see: [RFC 4303](#)] and/or Authentication Header (AH) [see: [RFC 4302](#)] and a set of cryptographic algorithms to be used by the SAs to protect the traffic that they carry. In this document, the term "suite" or "cryptographic suite" refers to a complete set of algorithms used to protect an SA (Security Association). An initiator proposes one or more suites by listing supported algorithms that can be combined into suites in a mix-and-match fashion. IKE can also negotiate use of IP Compression (IPComp) in connection with an ESP and/or AH SA. IKEv2 is described in [RFC 4306](#).

OCSP

The Online Certificate Status Protocol (OCSP) [RFC2560] defines a protocol for obtaining certificate status information from an online service. An OCSP responder may or may not be issued an OCSP responder certificate by the certification authority (CA) that issued the certificate whose status is being queried. An OCSP responder may provide pre-signed OCSP responses or may sign responses when queried. OCSP is described in [RFC 6277](#).

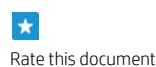
P4000 SAN

The P4000 Storage Area Network (SAN) Solution (SAN/iQ) protocol is the command line interface that is used to interface with the P4000 Storage Systems from the Hosting Device. The P4000 Command Line Interface (CLI) is installed with Insight Remote Support.

Note: The P4000 SAN Solution is sometimes referred to as CLiQ (or cliq), which is the name of the command used within the P4000 SAN Solution.

P6000 CV

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P6000 Command View (CV) is the storage management software used to monitor HP Enterprise Virtual Array (EVA) devices. Insight Remote Support uses ELMC to monitor the array controllers for new log entries and communicates this information back to the Hosting device. The Hosting Device communicates with P6000 CV over TCP port 2372 to query the software for configuration and event details.

RIBCL

Remote Insight Board Command Language is an Extensible Markup Language (XML) based command language for managing HP ProLiant Servers (series 300 and higher) via the Integrated Lights Out (iLO) interface. Insight RS uses RIBCL to communicate with the server onboard administrator (OA) to gather configuration information and event details for monitored devices. RIBCL communicates using HTTPS (TCP port 443).

SNMPv1

Simple Network Management Protocol version 1 is a protocol developed to manage nodes (servers, routers, switches, and hubs) on an IP network. SNMPv1 is described in [RFC 1157](#). SNMPv1 is an unencrypted communication service that communicates over UDP port 161. SNMPv1 is a simple request/response protocol (responses are not acknowledged). The Hosting device issues a request and a monitored device returns a response.

SNMPv2

Simple Network Management Protocol version 2 or more specifically, SNMPv2C (a subset of SNMPv2), is an extension of SNMPv1. It also is an unencrypted communication service that communicates over UDP port 161. SNMPv2 is described in [RFC 1441](#) and includes enhanced protocol operations to the SNMPv1 protocol that include the GetBulk operation (to retrieve large blocks of data) and the Inform operation (allowing one Network Management System to send trap information to another Network Management System and receive a response or acknowledgement). If Inform operation responses are not acknowledged, the SNMP agent will resend the Inform message.

SSH

The Secure Shell (SSH) protocol is an application-layer protocol which permits secure remote access over a network from one computer to another. SSH negotiates and establishes an encrypted, and authenticated connection between an SSH client and an SSH monitored server. SSH provides data integrity checks, prevents eavesdropping, and modification of sensitive data transferred between the Hosting Device and monitored systems. SSH typically uses TCP port 22, but alternative port numbers may be assigned to the SSH server. SSH is described in [RFC 4251](#).

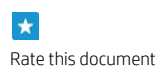
Although the SSH protocol is typically used to log into a remote machine and execute commands, it also supports tunneling, forwarding arbitrary TCP ports and X Windows System, version 11 (X11) connections. It can transfer files using the associated Secure File Transfer Protocol (SFTP) or Secure Containment Protocols (SCP).

The SSH protocol exists in two versions. Several security vulnerabilities have been identified in the original SSH protocol version 1, therefore it should be considered insecure and should not be used in a secure environment. Its successor, SSH protocol version 2, strengthened security by changing the protocol and adding Diffie-Hellman key exchange and strong integrity checking via message authentication codes. HP RDC and HP RDA use SSH protocol version 2 for most connections.

SSL and TLS

The Secure Sockets Layer (SSL) and Transport Layer Security (TLS) protocols are application-layer protocols which provide data encryption and authentication. TLS 1.0 is an updated version of SSL v3. SSL and TLS use X.509 certificates, also known as “digital” certificates, for authentication. Although most users are accustomed to working only with server certificates, SSL and TLS can be configured to require client-side certificates which provides password-less two-way authentication. The Hosting Device and monitored devices authenticate using X.509 certificates. Also, all communications between the client browsers and the Hosting Device are protected by SSL. The Remote Support Configuration Collector System supports both SSL V3 and TLS 1.X. These two protocols are most ubiquitous in HTTPS on TCP port 443. Other protocols and applications also utilize SSL and TLS for security.

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Telnet

Telecommunications Network (Telnet) is an application-layer protocol that was developed for providing remote terminal sessions. Some older storage devices, routers, switches, and other devices will support only telnet for network access. Although it is insecure, Insight Remote Support uses this protocol to provide support for these legacy devices. Telnet does not provide encrypted transport of data and is considered to be an insecure communication service. Today, most operating systems use SSH in place of telnet as the standard terminal communication protocol. Telnet is described in [RFC 854](#). Telnet has been assigned to TCP port 23, however it may be configured to run on other ports

WBEM

Web Based Enterprise Management (WBEM) is an initiative based on a set of management and Internet standard technologies developed by the Distributed Management Task Force (DMTF) to unify the management of enterprise computing environments. WBEM is really a collection of Internet standards and [DMTF open standards](#): Common Information Model (CIM) infrastructure and schema, CIM-XML, CIM operations over HTTP, and Web Services for Management (WS-Management). The Common Information Model (CIM) provides a common definition of management information for systems, networks, applications and services, and allows for vendor extensions. WS-Management is a specification of a SOAP-based protocol for the management of servers, devices, and applications. WBEM can be encapsulated inside either HTTP or HTTPS. HP Insight Remote Support does not support unencrypted WBEM communications. All Insight Remote Support WBEM traffic is encrypted using HTTPS on TCP port 5989.

Windows Management Instrumentation (WMI) is the Microsoft proprietary implementation of WBEM. WMI runs as a DCOM (Distributed Component Object Model) service which in turn uses RPC (Remote Procedure Call) and other associated DCOM services. The WMI Mapper is an application that provides a two way translation interface between DCOM and WBEM. WMI Mapper is required for any Windows monitored system supporting WBEM. Indications to be monitored by Insight Remote Support.

WS-MAN

WS-MAN or Web Services Management is a [DMTF open standard](#) defining a soap based protocol for the management of servers, devices and applications. HP Insight Remote Support uses WS-MAN to communicate with the Superdome 2 Onboard Administrator.

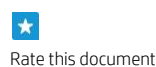
WMI

Windows Management Instrumentation (WMI) is Microsoft Corporations implementation of the Web Based Enterprise Management (WBEM) and Common Information Model (CIM) schema. WMI is a Windows API that can be leveraged to provide remote management and

Active Health System

HP Active Health System tracks configuration changes on ProLiant Gen8 servers with attached Smart Memory and Smart Drive devices, enabling you to eliminate time spent running diagnostics, reproducing problems, and describing errors to HP support engineers. Changes to the device configuration are reported to Insight RS using a secure (HTTPS) connection between the ProLiant Gen8 iLO4 (Integrated Lights Out) and the Hosting Device. HP Insight RS will package and forward the configuration changes to HP over a secure HTTPS connection. Active Health System information is not customer viewable.

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Insight Remote Support

HP Insight Remote Support version 7.0.8 stores information in specific locations on the Hosting Device. Permissions on these directories are set to deny access to all users except Hosting Device System Administrators and the Windows System account. The Installer can change the default locations for these directories during installation. The standard (default) locations for Insight Remote Support are as follows:

- Data %SystemDrive%\ProgramData\HP\RS\DATA
- Log Files %SystemDrive%\ProgramData\HP\RS\LOG
- Configuration Files %SystemDrive%\ProgramData\HP\RS\CONFIG
- Executable Files (32-bit) %SystemDrive%\Program Files\HP\RS
- Executable Files (64-bit) %SystemDrive%\Program Files (X86)\HP\RS

Event Management

Insight Remote Support relies on the onboard diagnostic monitors to detect hardware events on monitored devices. When events are detected, notification is sent to the Hosting Device (and any other monitoring host) via one of the management protocols listed in Table 1 above. The management protocol used is determined by platform and policy. Insight Remote Support supports platforms (refer to the [Monitored Devices Configuration Guide](#) to determine the supported protocol for your device).

When the Hosting Device receives an event from the managed device, the Insight Remote Support software on the Hosting Device will screen the event to determine whether or not the event may require action by HP to address. If the analysis determines that action by HP may be necessary, the event will be packaged with the contact information for the affected device stored in Insight Remote Support and sent to HP via HTTPS (TCP/443). Event details are typically stored locally on the Hosting Device for 24 to 36 hours after analysis and can be viewed at the following (default) location:

Event Data
%SystemDrive%\ProgramData\HP\RS\DATA\ANALYSIS\attachments\[filename].xml

Data Collections

Insight Remote Support collects configuration information about devices in your environment. This data is used to aid in restoring your device to production status. Depending on your support agreement with HP, it can also be analyzed and compared with information in HP's knowledge database to provide recommendations to improve performance, or to avoid potential unwanted downtime. Data is collected using management agents (like WBEM) to query the device and report data back to the Hosting Device. This information is packaged by the Hosting Device and sent to HP via HTTPS (TCP/443). Data collections are compressed and stored locally on the Hosting Device for varying lengths of time depending on the collection type and schedule. Stored collections can be viewed at the following (default) location:

Collection Data:
%SystemDrive%\ProgramData\HP\RS\DATA\collection\results\[Collection_ID]\[SubcollectionID]\[filename].zip

The default collection retention policies for on demand (RunNow) and scheduled collections are shown in Table 3, the Number Retained value indicates the maximum number of most recent collections that will be stored locally on the Hosting Device.

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Table 3: Data Collection Retention Default Schedule

Collection Name	Default Collection Schedule	Number Retained for 'RunNow' Collections	Number Retained for 'Scheduled' Collections
ActiveHealthServiceCollection	Weekly	1	2
MetricsCollection	Weekly	7	4
NetworkConfigurationCollection	Weekly	2	3
P4000FamilyConfigurationCollection	Daily	2	5
PerformanceDataCollection	RunNow Only	2	N/A
SANConfigurationCollection	Weekly	2	3
ServerBasicConfigurationCollection	Monthly	2	3
StorageConfigurationCollection	Weekly	2	3
SupportDataCollection	RunNow Only	1	N/A
vCenterApplicationDataCollection	Weekly	1	2

Logging

The Hosting Device keeps a record of Insight Remote Support activities in the following (default) location:

Log Data:
C:\ProgramData\HP\RS\LOG\{Log_Name}.log

Data Sent to HP

This section describes Insight Remote Support data sent from a Hosting Device to HP. Some ProLiant Gen8 and c-Class BladeServers have the ability to send data directly to HP. For these devices, refer to the HP Insight Online Direct Connect Architecture and Security Model in the Whitepapers section of the [Insight Remote Support Information Library](#)

Data sent to HP from the Hosting Device can be sent to HP directly or via a proxy server. If a proxy server is used, the proxy settings are configured using the Insight Remote Support User Interface: Administrator Settings → Settings tab. If a proxy username and password are required, the password is encrypted and stored in a binary file on disk. If the proxy username and password are changed at the proxy, they must also be changed in the Insight RS Console to ensure connectivity to HP is uninterrupted. All transport sessions to HP are encrypted using SSLv3/TLS over HTTPS (see note below). Connections are always initiated by the Hosting Device outbound to HP and are authenticated using X.509 Digital Certificates and a Global Unique Identifier (GUID) that is unique to the Hosting Device. All data sent to HP is via a HTTPS connection to a single destination URL (<https://services.isee.hp.com>). This destination is a virtual IP address that is automatically routed to an active server in one of the HP Corporate Data Centers (see figure 2).

Note: Insight Remote Support will allow SSLv2 connections from Monitored Devices to ensure compatibility with some platforms. All connections to HP require strong encryption (SSLv3/TLSv1.0 or higher) to ensure the best possible security during the transport of event and collection data to HP.

Data Sent to HP contains configuration information about devices in your environment. This information can be viewed using HP Insight Online. This may include diagnostic sense information, firmware information, model number, serial

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number, and other configuration data. Due to the nature of the configuration collection utilities, some potentially sensitive configuration details may be collected and sent to HP as part of the event or data collection. This could include IP Address, Fully Qualified Domain Name, MAC address, DNS Configuration, and Windows Domain Details. HP treats all collection data as HP Confidential while at HP. Access to this information is restricted to authorized HP personnel with a valid business reason for accessing this information. Device Administrator contact details such as system administrator name, phone number, and email address will also be added to the event or collection data prior to transport to HP. This is done to ensure HP has the necessary contact information in case a response from HP is required to affect a repair or to recommend a configuration change to avoid potential downtime.

All information collected by Insight Remote Support and sent to HP is used in accordance with the Insight Remote Support Terms and Conditions (see note below) and the [HP Online Privacy Statement](#).

Note: *For receiving remote support:*

Installing HP Insight Remote Support configures your IT devices being remotely supported to securely send support or service events, IT configuration information, diagnostic, configuration, and telemetry information to HP, together with your support contact information. No other business information is collected and the data is managed according to the HP Data Privacy policy.

To provide you advisories to optimize your IT environment:

If you choose to 'Opt-In' to be contacted by HP or your HP authorized reseller to optimize your IT environment, HP or HP authorized resellers may use the collected configuration data to provide you with recommendations, sell or deliver solutions, to optimize your IT environment. These providers may be located in other countries than your HP IT hardware locations. HP's providers are required to keep confidential information received from HP and may use it only for the purpose of providing advisories and recommendations on behalf of HP. You will have the option to specify your HP authorized reseller(s) or support provider(s) during setup of HP Insight Remote Support software. Only the HP authorized resellers and support providers you associate with your devices can receive your configuration data to individually contact you for making IT environment recommendations, sell, or deliver solutions.

Automated Connections to HP

Insight Remote Support will automatically open a HTTPS communication channel to HP for the submission of service events, data collections and automatic device registrations. In addition to these messaging events, HP Insight Remote Support Client installations will send 'Heartbeat' messages to the HP Data Center once every 24 hours to verify connectivity. Insight Remote Support Heartbeats are used to verify that communication with HP is functioning properly. Missing 'Heartbeats' are only acted upon for customers where this is a contractual deliverable. Currently, this is limited to customers who have purchased either a Mission Critical or a Mission Critical Partnership contract. If there are open service events or pending data collections, the Insight Remote Support Client will automatically connect to the HP Data Center every 10 minutes to check for status updates or to confirm the successful submission of pending data collections. If there are no open service events or pending data collections the Insight Remote Support client will connect to HP every 6 hours to check for and retrieve routine messages and updates.

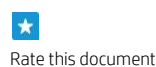
Connection Retries

If an Internet connection fails to connect to HP, it will automatically retry the connection after two minutes. If the connection still fails, it will retry again after four minutes and again after eight minutes. Doubling the time after each failed attempt until the maximum number of retries (10 retries or 2048 minutes) is reached. If all connection attempts fail, the data will be discarded and an Application Failure message will be displayed in the Insight Remote Support console.

Email Notifications

Insight RS has the capability of sending email notifications to the default and device contacts when certain events occur. You can enable email notification on the Administrator Settings → Integration Adapters tab, enabling email notification in the Insight RS Console, allowing you to receive email notification for any or all of the following events:

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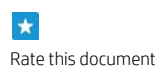


Notification State	Description
Case Opened	Default and backup contacts notified when a case is opened in the HP data center. Note that service events generated by test events are never opened so an email will not be sent for test events.
Case Closed	Default and Backup contacts notified when a case is closed in HP data center. Emails are also sent for service events generated by test events.
Collection Sent	Default and Backup contacts notified each time data collected about a device is sent to HP.
Application Failure	Default contact notified when the Insight Remote Support application fails, or when a data transport failure occurs.
Software Management Updates	Default contact notified whenever there is a new software update is available.
Entitlement Expiration	Default and Backup contacts notified when a warranty or contract is about to expire. Notifications are sent at 90, 60, 30 and 0 days prior to expiration.
New Device Discovered	Default and Backup contacts notified when a new device has been discovered.
Hosting Device Threshold Exceeds %	Default and backup contacts notified when the Hosting Device's capacity exceeds the specified percentage of devices that Insight RS can support.

Email messages may contain device IP Address and Fully Qualified Domain Name. This information is sent from the Hosting Device via unencrypted email.

HP does not recommend sending unencrypted email notification messages to destinations outside of your company.

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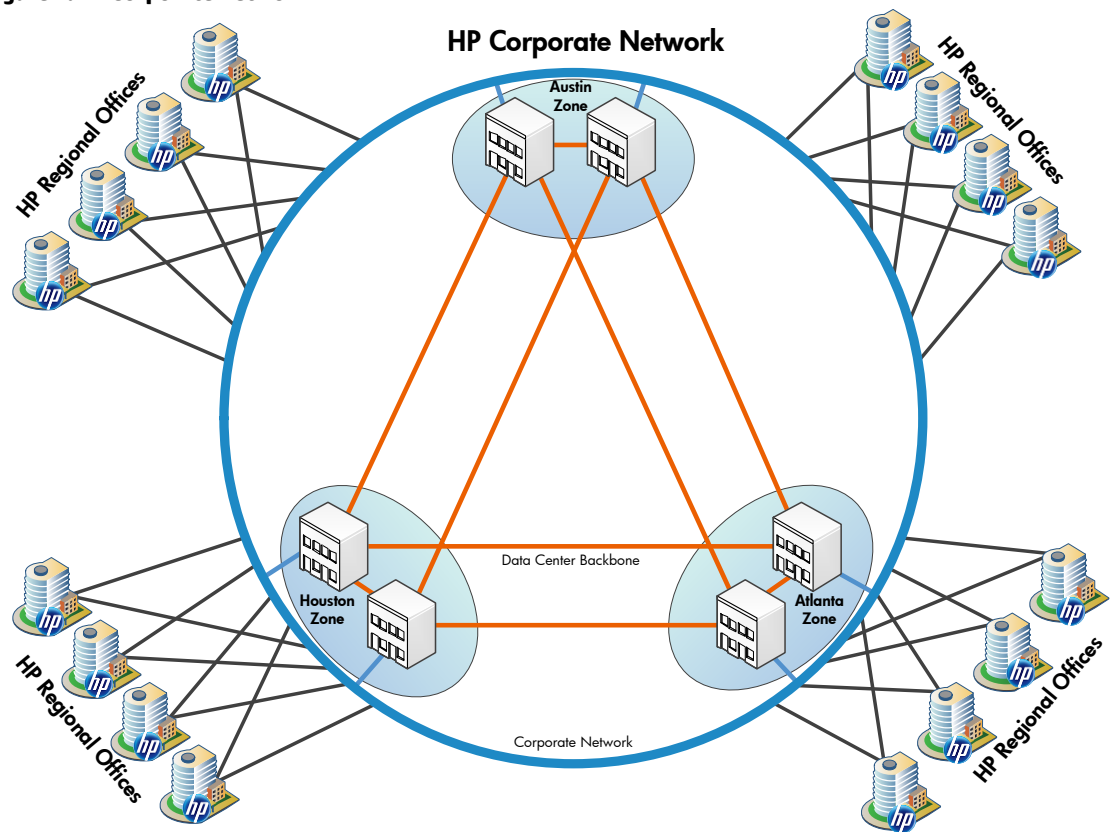
Insight Remote Support at HP

HP Data Centers

All customer data received by HP is treated as “HP Confidential” and treated in accordance with HP’s Data Handling guidelines for HP Confidential information.

Customer data is stored in one of six HP Global IT Next Generation Data Centers (NGDC) — two each in the geographical zones of Austin, Texas; Houston, Texas; and Atlanta, Georgia — that have site-to-site and zone-to-zone business continuity and disaster recovery capabilities. The NGDCs operate continuously (24x7) in a lights-out computing environment with strict physical and logical access control mechanisms. HP corporate data centers are concurrently maintainable and are designed to meet or exceed the American National Standards Institute / Technology Industries Association (ANSI/TIA) 942-2005 Tier III Data Center standards.

Figure 2: HP Corporate Network



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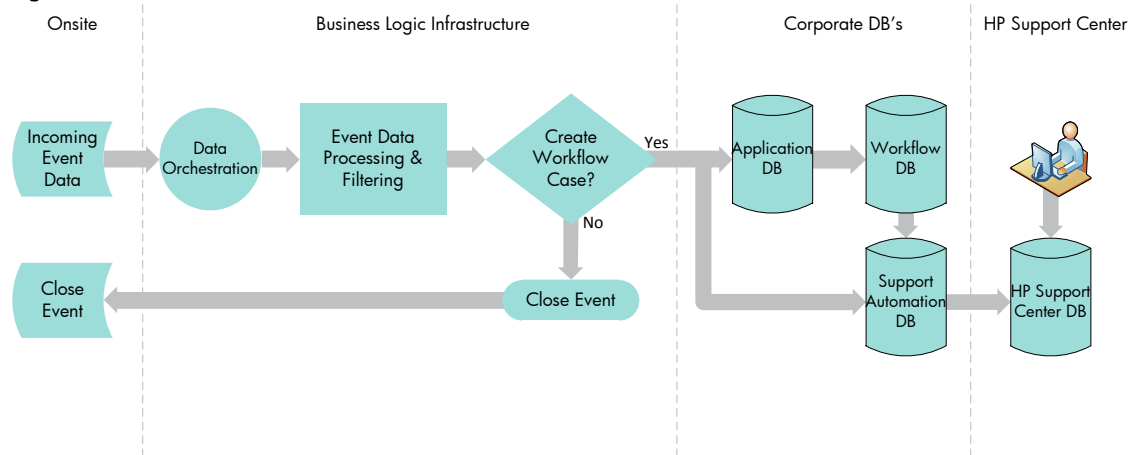


Data Stored at HP

Data transmitted to HP is received by the Application Servers in the Remote Support Data Center (RSDC). The Application Servers are responsible for the initial processing of data and routing it to the appropriate destination. There are two basic types of data transmitted to HP from Insight RS: event data and collection data. Event data is any data that is the result of a hardware or software event that occurs on a monitored device that is sent to HP. Collection data contains configuration details about monitored devices in your enterprise. Configuration information can be used to restore device configuration parameters after a hardware component has been replaced. It can also be used for configuration analysis. Configuration analysis compares your device configuration information against HP's known problem database in order to identify potential configuration issues that could impact production and/or performance.

Note: The RSDC servers support Global Server Load Balancing (GSLB) and Site-to-Site failover, but have not implemented Zone-to-Zone failover.

Figure 3: Event Data Flow at HP



Data Orchestration

When Insight Remote Support event data is received at HP, the first step is to determine the type of data coming in and route it to the correct parsing engine. Event data is forwarded to the event processing engine and collection data is forwarded to the collection processing engine.

Event Processing

Every device monitored by Insight Remote Support is assigned a unique identifier call a Global Device Identifier (GDID). The event processing looks at the GDID in the event to determine if there is a record for this device in the Application Database. Event processing will also parse the data so it can be analyzed.

Event Filtering

Event Filtering uses smart analytics to determine whether or not an event requires action by HP. This is done by comparing the event to a rules engine to determine if the event meets all of the requirements necessary to open a service request.

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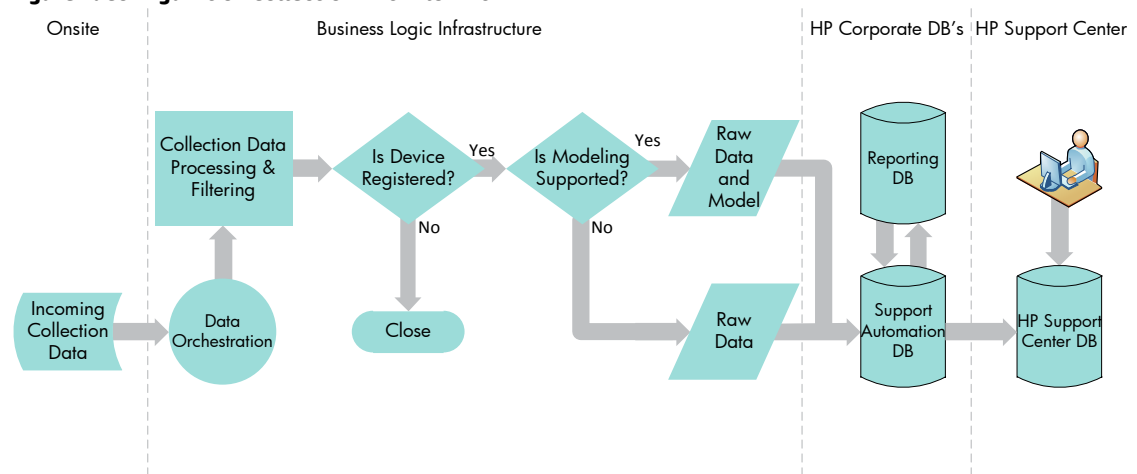
Entitlement

If an event passes the initial event filtering process, it will go through entitlement analysis (denoted by the ‘Create Workflow Case?’ decision box above). Entitlement analysis checks the device entitlement parameters (serial number, model number, contract identifier) against the HP entitlement database to determine the appropriate Service Level Agreement (SLA) for the device. If a valid entitlement is found, the event is forwarded to Event Correlation. If no valid entitlement is found, the event is closed with a status of no entitlement.

Event Correlation

Once an event is entitled, it goes through one final correlation check. All currently open service requests for this device are checked to determine if the event matches the description of an open service request. If no match is found, a new service request is opened in the workflow system and routed to the appropriate support team for resolution. If a match is found, the event is marked as a duplicate event and correlated with the open workflow case.

Figure 4: Configuration Collection Data Flow at HP



Collection Processing

Collection data, like event data, is parsed to obtain the device GDID and entitlement information. The GDID is used to identify the device for which the collection information originated. The entitlement information is used to determine if the device is covered under a support agreement that authorizes collection information to be stored and analyzed by HP. If collection processing determines that collections are supported for this device's GDID, the collection data will be sent to the Support Automation Database (SADB).

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HP Insight Online

HP Support Center

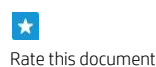
HP Insight Online is a new capability with Insight Remote Support version 7.0. It is a cloud-based IT Management and support solution. HP Insight Online lets you provision, monitor, and remotely support devices in your enterprise from a single online portal. Data collected from your devices can be viewed online using HP Support Center. HP Insight Online allows customers (and optionally, HP Authorized Resellers and Authorized Support Providers) with Insight Remote Support version 7.0 or later to monitor the status and support details of devices in their enterprise. Enabling HP Insight Online can be done from the onsite Insight Remote Support Setup Wizard by checking the Optional Settings box “View your IT environment data via HP Insight Online” on the Register Hosting Device page and entering your HP Passport user account information. Insight RS Administrators can also enable or disable HP Insight Online by selecting “Administrator Settings” from the drop-down menu and clicking on the “settings” tab. To enable HP Insight Online, check the box “View Data in HP Support Center”. To disable HP Insight Online, uncheck the box. This will prevent further device configuration data from being sent to HP. Administrators can verify access to HP Support Center from the Insight RS Console by entering their HP Passport Username and Password in the setup wizard or on the Administrator Settings tab. The HP Passport username is retained in the Insight RS Console settings; however the HP Passport password is passed to HP Passport for authentication only and is not retained in the Insight RS Console.

HP Passport

Access authentication for HP Insight Online is managed by HP Passport.

HP Passport maintains access information for most HP online applications, HP Passport stores basic personal information (e.g., user id, password, name, e-mail address, country, and language preferences) in an encrypted database. This information is managed according to HP’s strict privacy policies. These policies have been reviewed by TRUSTe and have received the “TRUSTed Web Privacy – Cloud Certification”.

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Remote Device Access (RDA)

HP offers several options for establishing a secure connection between HP and your network, allowing an HP support specialist—with your authorization—to remotely access your monitored systems and devices. Using HP RDA, an HP support specialist can login to your system, observing normal security processes and procedures in order to provide remote hardware or software support for faster resolution of problems.

HP Remote Access can be setup up on demand (ad hoc), or preconfigured (entitled) prior to use.

Ad Hoc

Ad Hoc connections can be used if there is no pre-configured solution installed, or if your security policy does not allow static inbound Business-To-Business (B2B) access connections into your corporate network. In the ad hoc solution, the customer and HP agree to engage in an immediate RDA session. This connection type allows for the creation of an ad hoc, or spontaneous, remote connection to your desktop using lightweight applications such as HP Virtual Support Room (VSR) or the HP Instant Customer Access Server (iCAS). Once you share your desktop within the Virtual Support Room, or allow HP to connect via the iCAS, the support engineer can leverage this connection to provide access to target systems inside your corporate network. This solution must be initiated from a system connected to your corporate network.

Ad Hoc RDA options include:

- HP Virtual Support Rooms (VSR) – A web-based desktop sharing application
- HP Instant Customer Access Server (iCAS) – A meet-in-the-middle access model that allows HP remote access connections between HP and a customer network using Secure Shell (SSH) tunneled over an HTTPS connection

Entitled

Entitled Remote Device Access describes a connection solution which must be deployed and configured at your site before support can be delivered (this is sometimes called a pre-configured solution). This may include routers or other hardware specifically configured to allow connections from HP. This connection type allows a support engineer to access a pre-configured Customer Access System (CAS) within your corporate network to gain access to HP supported systems and devices. HP can initiate an entitled connection at any time with your consent, but without requiring your assistance to establish the connection.

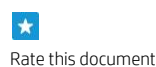
Entitled Remote Access options include:

- SSH-Direct – The SSH tunnel runs bare over the Internet
- IPSec VPN Connectivity – The SSH tunnel runs over a peer-to-peer IPSec VPN tunnel between HP and your company network
- SSL VPN Connectivity – This solution requires a SSL VPN concentrator on your network to be configured to allow access for HP Support. Connections are tunneled through a secure SSL (HTTPS) connection over the Internet.
- Integrated Services Digital Network (ISDN) Connectivity* – The SSH tunnel runs over an ISDN connection

***Note:** The ISDN option is only available in select countries.

Most of the Entitled Remote Access solutions leverage the end-to-end encryption and application tunneling capabilities of SSHv2. While using SSHv2 is strongly recommended, some versions of Entitled Remote Access can be configured without SSHv2. Not using SSHv2 can reduce the security profile and limit the functionality of the RDA solution.

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Service Value

The RDA solution provides HP customers an information security compliance level so that customers will meet most government and industry regulations. Authentication, access control, and secure communications conform to industry best practices.

Authentication

Customers can identify that they are securely connected to HP support specialists. Only authorized HP support specialists are able to establish connections, authenticated with digital certificates.

Access Control Overview

HP customers using RDA have full control of all incoming connections. Authorization and access restrictions can be configured to meet the customer's own security needs. For unattended RDA, audit trails are stored in audit log files.

Secure Communications

All communications meet current security best practice standards on encryption. Multiple layers of security ensure that HP customers can use RDA with confidence.

Remote Device Access Using SSH

All unattended RDA solutions rely on an SSH (SSH-2 protocol) tunnel running between the support specialist's desktop and a designated Customer Access System (CAS) deployed either in the customer Demilitarized Zone (DMZ) or on a trusted network.

An SSH server is required on the customer network acting as a Customer Access System (see CAS below). A SSH client is typically used for establishing connections to a SSH server accepting remote connections. An SSH server is commonly present on most modern operating systems, including Microsoft Windows, Mac OS X, Linux, FreeBSD, HP-UX, Tru64 UNIX, and OpenVMS. Proprietary, freeware, and open source versions of SSH client are available with various levels of complexity and functionality.

Most SSH implementations can be configured to comply with customers' security policies. For example:

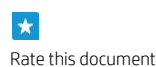
- The protocol can be limited to SSH-2 only
- Selection of encryption algorithm (3DES, AES, AES-256, etc)
- Allow only private/public key authentication (disallow password authentication)
- Use SecurID and other token-based authentication methods

Additionally some implementations support the use of X.509 certificates (also called an HP DigitalBadge) and two-factor authentication.

Customer Access System (CAS)

A Customer Access System (CAS) is required for all unattended RDA methods. By hosting the SSH server, the CAS provides a central point for customers to control remote access into their environment. Customers determine the login of each HP user individually to allow or deny specific services or access to specific computers within their network. The HP SIM Central Management Server (CMS) or the Insight RS Hosting Device used by the HP Insight Remote Support Solution can also function as a CAS.

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A CAS may be implemented on any customer-owned system capable of running a compatible SSH server. HP also offers a self-contained virtualized CAS solution.

Customer-owned CAS

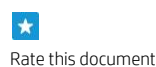
The customer may choose to provide their own CAS. The primary requirement is a functional SSH server such as OpenSSH. Microsoft Windows, Linux, HP-UX, OpenVMS, and Tru64 UNIX operating systems may be used. HP recommends that the customer configure SSH to accept only protocol version 2 and strong encryption (that is, AES (Advanced Encryption Standard), Triple-DES (Data Encryption Standard), or AES-256). Firewalls should also be configured to allow SSH access only from HP's access servers.

Virtual CAS

The Virtual CAS is provided by HP for free and is the HP preferred method for customers installing CAS functionality within their network. The Virtual CAS provides enhanced security and management functionality. It is a software-only solution based on a VMware image of a virtual machine running Ubuntu Server. Virtual CAS features include:

- Runs on VMware Server ESX; ESXi or Oracle VM VirtualBox
- It can run on the Hosting Device of the HP Insight Remote Support 7.X solution
- Based on open source software
- An easy-to-use administration web interface
- Implements SSH authentication using X.509 certificates
- The authentication is compatible with HP's VeriSign-administered internal Public Key Infrastructure (PKI) (known internally as HP DigitalBadge)
- Certificate Revocation List (CRL) access is available either via file or Online Certificate Status Protocol (OCSP)
- Fine-granularity access control – customers can specify user level access to targets including TCP ports
- Easy-to-use software update mechanism based on apt-get. The virtual CAS will poll the HP Advanced Packaging Tool repository for software updates and security patches. The customer has full control on how and when these updates may be applied to the Virtual CAS
- Can be used with SSH-Direct or IPSec VPN solutions

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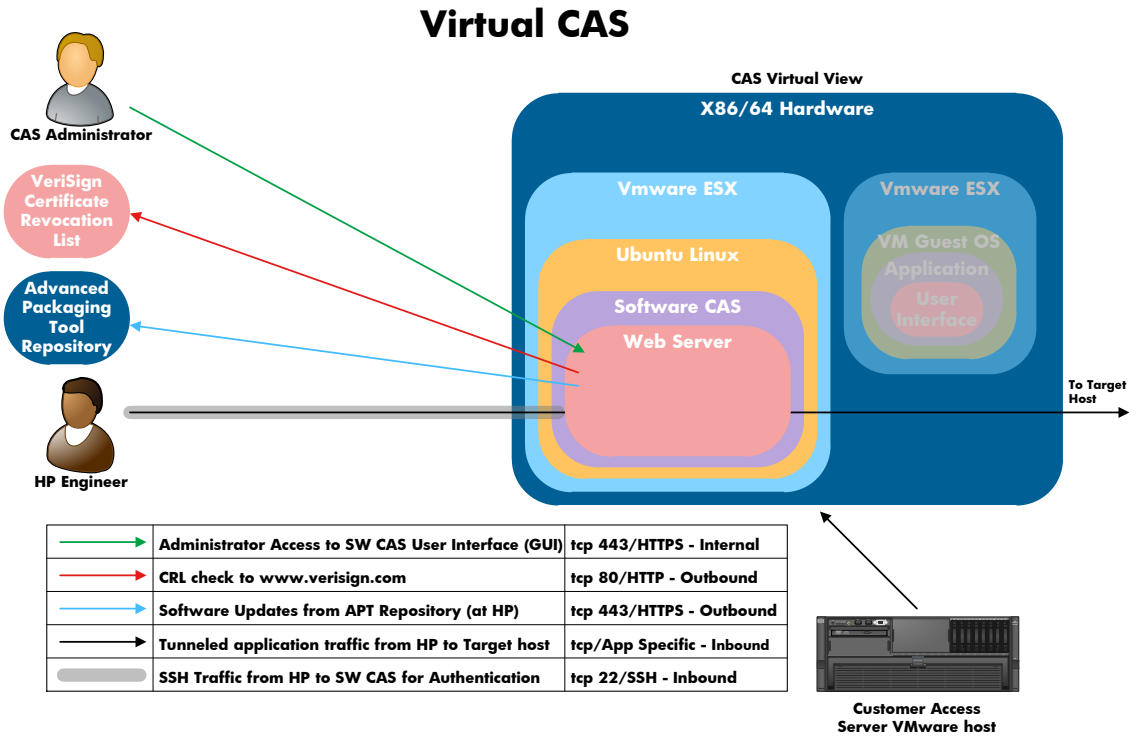
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Figure 5: Virtual CAS



HP Instant Customer Access Server (iCAS)

HP Instant Customer Access Server (iCAS) is a lightweight connection tool that allows an HP support agent to quickly and securely connect to a customer's environment to aid in diagnosis and repair of supported hardware devices. The customer runs the iCAS software as a browser plug-in on any Windows or Linux computer which has network access to the device the HP support engineer is attempting to access. HP iCAS uses a meet-in-the-middle connection paradigm to facilitate the remote access session by establishing a tunneled SSH session to a Remote Access Meeting Server (RAMS). The HP engineer generates a unique connection key that is used to couple the HP Engineer and Customer SSH connections together creating an end-to-end SSH tunnel between the HP Support engineer desktop and the iCAS host. Once the session key is exchanged, the session is established as follows:

- HTTPS connection occurs (using TCP/443) from iCAS host to RAMS using URL and Session key provided by HP Support engineer
- Customer's SSH connection (using TCP/ 2022) is tunneled via HTTPS to RAMS Server
- The HP engineer session sees the customer session connected to the RAMS
- An HTTPS connection is made from HP engineer browser to the RAMS
- The HP engineer's SSH connection (using TCP/2022) is tunneled via HTTPS to RAMS
- The unique session key insures that both sessions rendezvous on the RAMS and create a secure SSH tunnel

From this point the HP engineer can request access to the affected system in the customer network by tunneling through the SSH tunnel (inside the HTTPS tunnel) to the target device inside the customer network. The customer must specifically grant access and provide the access credentials to the HP engineer before the connection to the target device can be established.

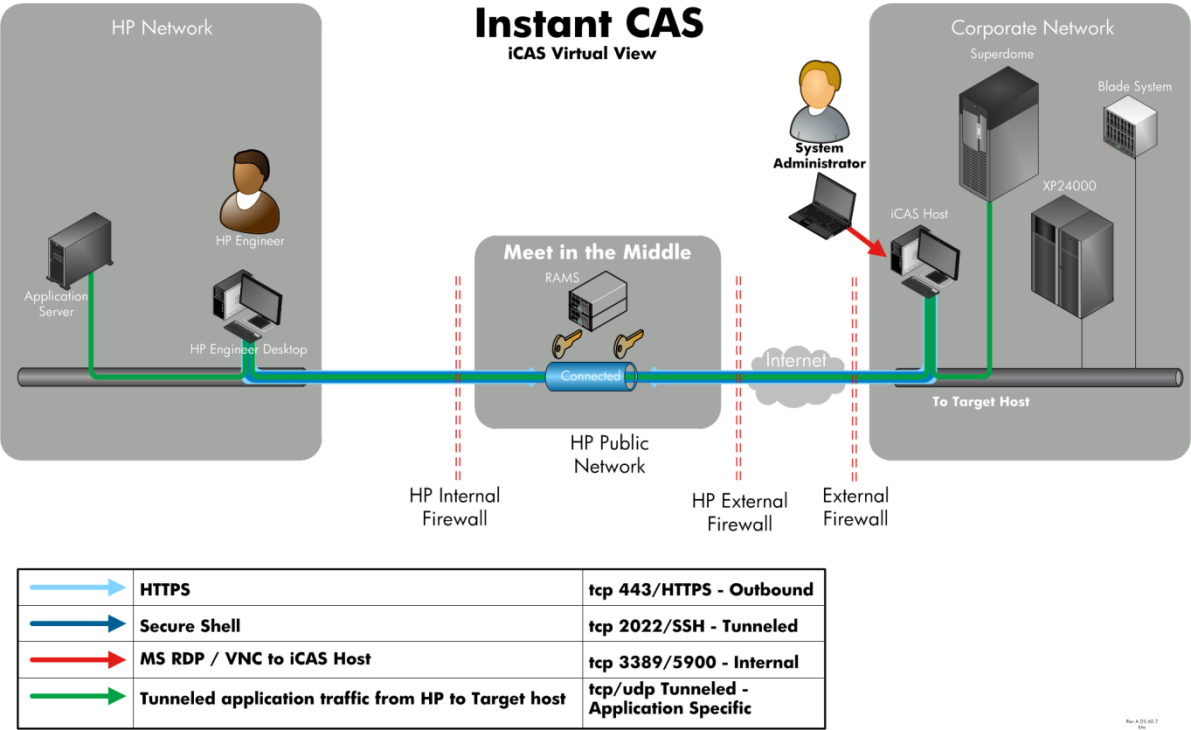
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Figure 6: Instant CAS (iCAS)



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RDA Access Controls

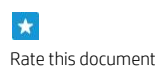
Access Controls at HP

HP manages all remote access customers in an internal portal called Remote Access Portal (RAP). Customer information and their connection data are centrally and securely managed via this central portal. Each customer can be associated with individual access rights so that narrow access permissions for this customer can be enforced, matching your security and access permission needs. The Remote Connectivity Database is the central place where the configuration data and access permissions are stored and encrypted in a secure HP Data Center facility.

An HP Support specialist must authenticate to the HP Remote Data Access (RDA) Infrastructure (Remote Access Portal System - RAPS, Remote Connectivity Toolbox System - RCTS, and Remote Access Connection System - RACS) using his or her HP-issued X.509 digital certificate, internally called Class A DigitalBadge, that employ two-factor authentication. The HP support specialist must have a physical ActivKey or ActivCard which is enabled by a password or passphrase. This is a physical handheld token issued to appropriate HP support personnel and issuance is controlled by HP business and security policies.

An HP support specialist must be granted permission to access a customer in RAP before they can see the connection details necessary to initiate a remote access session to a CAS on a customer network. If they are not able to see the connection details, they must contact the HP account owner and request access to the customer network in RAP.

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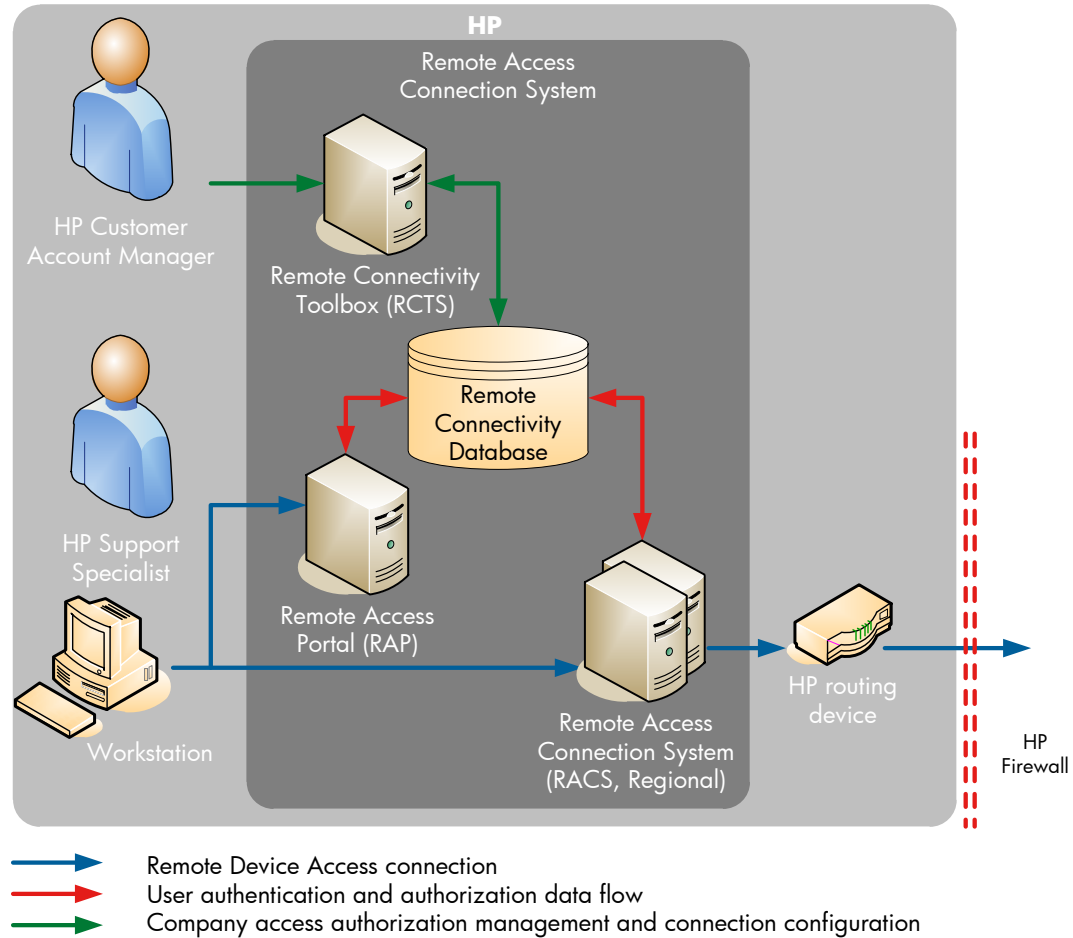
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Figure 7: Remote Access Connection System Details



A Remote Access Connection System (RACS) is an SSH server that can forward an SSH connection to an appropriate CAS. When the HP support specialist connects and is authenticated to the RACS, the SSH server on the RACS checks the security token issued by the RAP to ensure that the support specialist is allowed to connect to the customer's IP address. Upon successful authorization, the RACS will forward the SSH connection to the HP routing device. RACS servers are located in various HP data center locations.

Access Controls Onsite

For a primary defense, the customer external firewall can be configured to allow only RACS systems at HP to access their VPN devices and/or CASii. Although standard passwords can be used, it is recommended to configure SSH public/private keys instead. Some versions of SSH servers can be configured to use HP's DigitalBadge certificates for authentication. HP recommends that customers use the HP provided Virtual CAS, as this provides richer access control for customers.

One-time password systems, such as RSA's SecurID, can also be used if the customer's SSH server supports them.

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The CAS itself provides the second layer of defense. Depending on the CAS type, customers can define named employees, target systems, or even ports that HP support specialists are allowed to connect to.

The customer owns the security policies and access control into his/her environment and can specifically restrict connections to named HP support personnel and can terminate connections as needed.

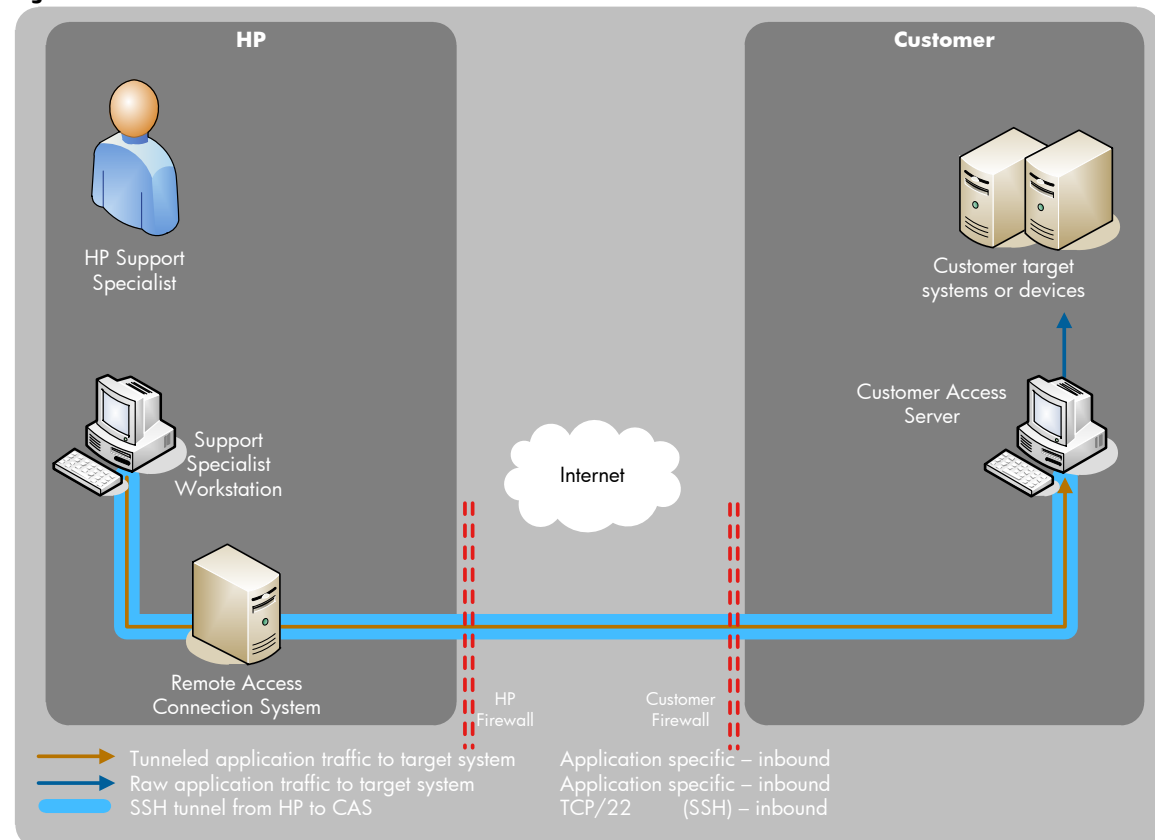
The HP Support specialist is also subject to the customer's own access control and security policies in that the customer must provide login credentials if needed for the device that HP wishes to connect to. For example if the HP support engineer wishes to logon to a UNIX server within the customer's network, the customer provides the logon name and controls what activities the HP support agent can perform. In this way, the customer oversees who from HP connects to their network and then controls where they can go and what they are allowed to do.

The third layer is the login credentials on the target system that must be known by the HP support specialist, typically pre-shared or shared on demand by the customer to HP over a different secure communication channel.

Connectivity Method: SSH-Direct – Secure Shell over Internet

The direct SSH option provides a simple and easy unattended RDA solution. The customer need only provide HP with an Internet Routable IP address for the CAS and allow one of the HP access servers to access it on port 22. The SSH-2 protocol is considered as secure as SSL.

Figure 8: SSH Direct



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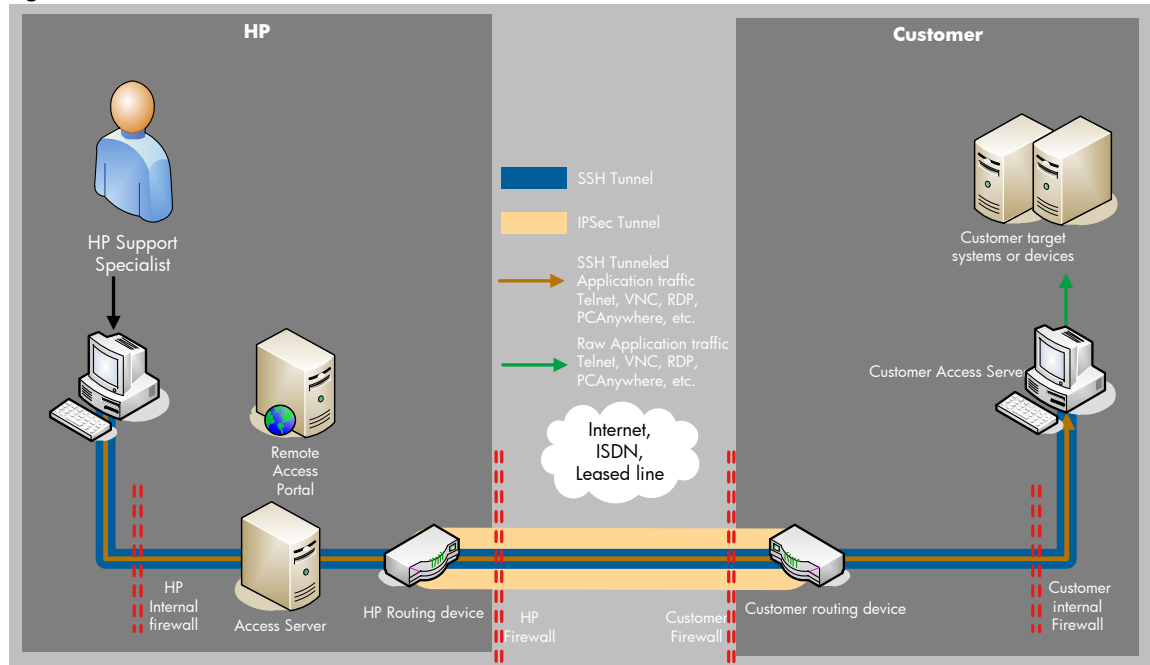
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Connectivity Methods for VPN Solutions

Many customers' security policies require that all inbound connections be protected inside a VPN connection that is terminated in their DMZ. HP offers a site-to-site IPSec VPN access solution for unattended RDA. SSH port-forwarding is still used, except that it is tunneled over IPSec using VPN routers. The combination of SSH and IPSec provides enhanced Internet security. SSH is recommended as it provides better end-to-end security as well as enhanced functionality (file transfer capabilities and application tunneling), but HP recognizes that this may not fit all security policies. Therefore, we offer site-to-site IPSec VPN connectivity with and without SSH tunneling. The following figures show both options.

Figure 9: General IPSec VPN Access with SSH



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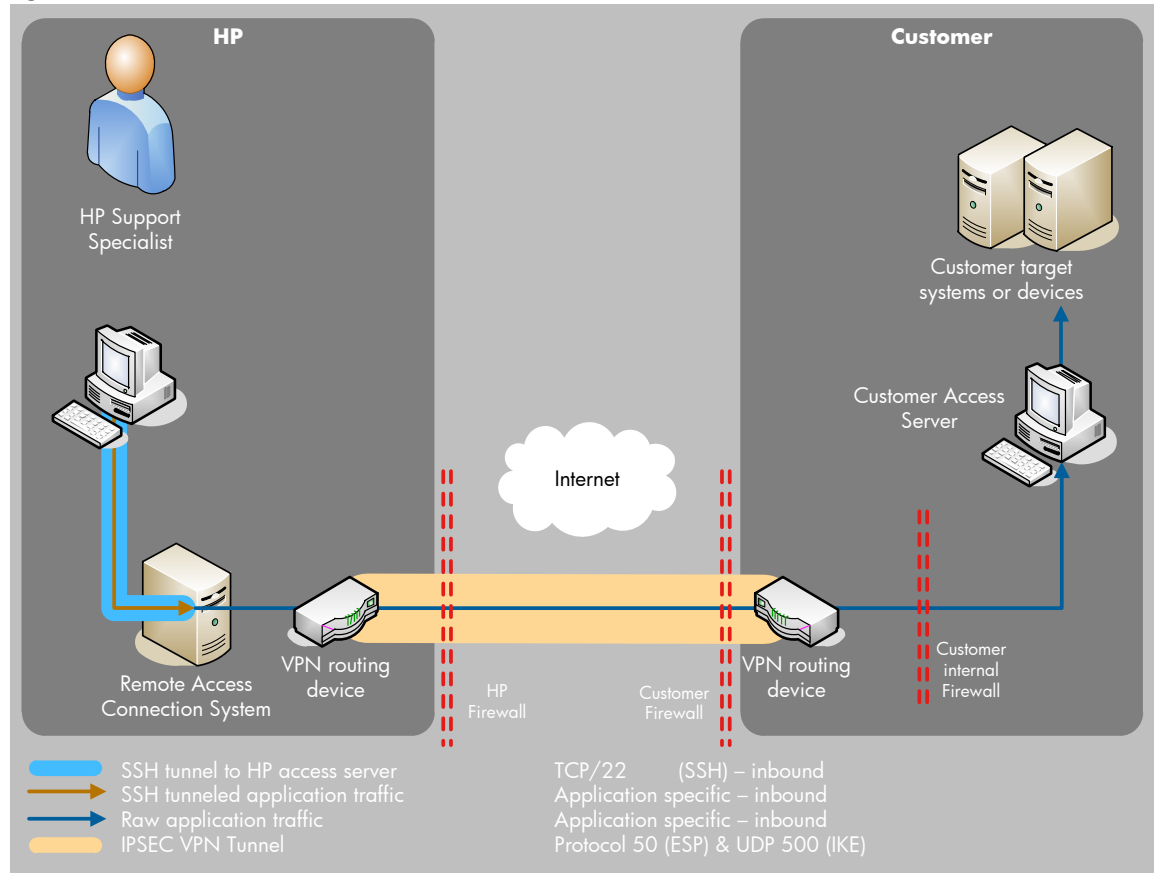
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Figure 10: General IPsec VPN Access Without SSH



IPSec VPN

With IPsec VPN, HP establishes an IPsec VPN with a customer-managed VPN device. HP's RDA VPN routers are successfully inter-operating with ProCurve, Cisco IOS, Cisco PIX, Check Point, Juniper, Linux and other IPsec VPN-capable devices at customer sites. IPsec VPN connections can be configured according to a customer's unique configuration requirements. With the IPsec VPN remote access option, the customer's network administrators are responsible for configuring and maintaining the IPsec configuration on their end of this B2B connection. HP is responsible for maintaining the HP side configuration. HP Support specialists can assist with troubleshooting any connection issues with this solution.

Connectivity Method for Integrated Service Digital Network (ISDN)

In some countries, HP offers the option of ISDN connectivity. As with VPN solutions, SSH port-forwarding is used over ISDN to provide secure remote access.

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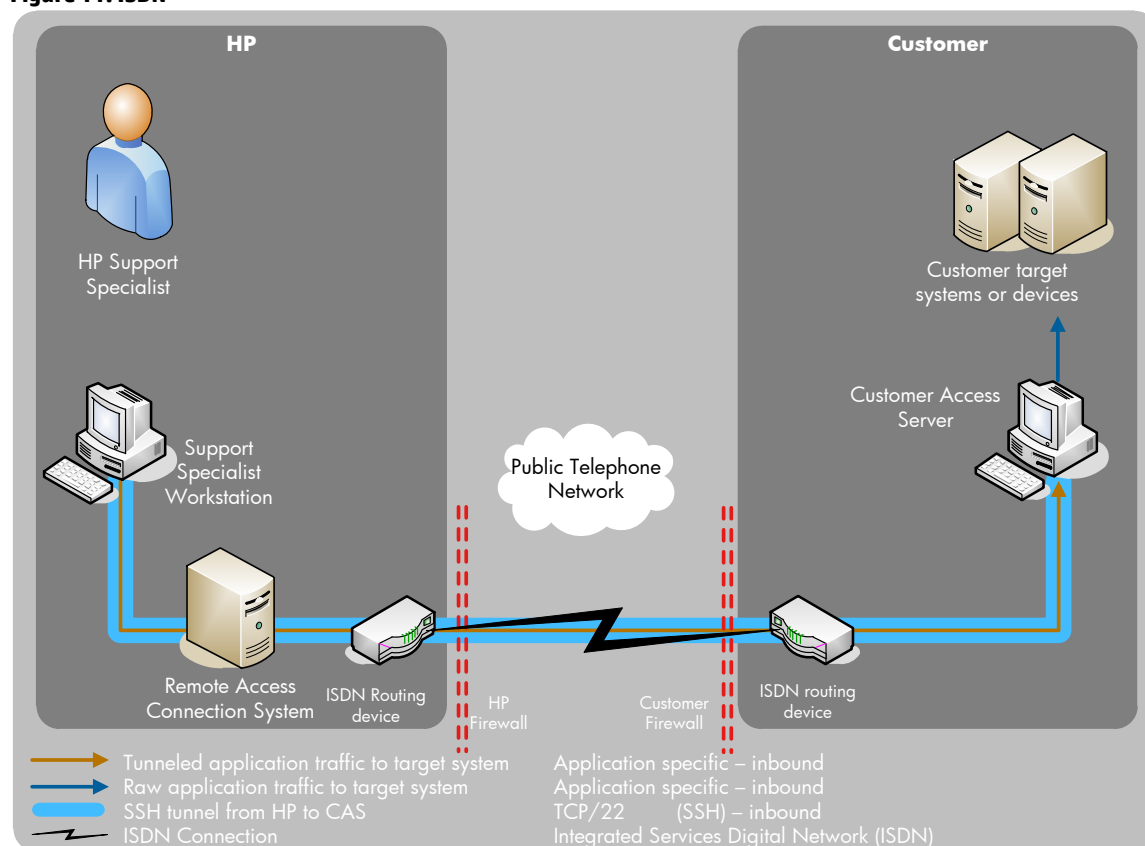
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Figure 11: ISDN



Attended RDA via Virtual Support Room

Virtual Support Room (VSR) is a lightweight, web-hosted meeting place that enables HP support specialists to connect to a customer enterprise covered under warranty or contractual agreement. Attended RDA is an ad-hoc connection method that can be used without any complex configuration or hardware setup.

VSR is based on HP Virtual Rooms and offers web collaboration functionality such as desktop sharing, file transfer, and desktop control. Like a real private meeting room securely locked with doors, the HP Virtual Support Room is a secure private protected online meeting place for two or more meeting participants.

The VSR meeting session involves two or more users virtually meeting in a Virtual Support Room and sharing a desktop for collaboration purposes. The collaboration session is initiated by the HP support specialist.

The HP support specialist will generate room keys for the Virtual Support Room and share those keys via unencrypted email or phone with the customer. The keys are required to enter the Virtual Support Room. The room keys are valid for one hour and must be re-generated after that time.

Joining a VSR session is a single mouse click action. A customer does not need more than a web browser, connecting via HTTPS to the HP Virtual Support Rooms infrastructure. The VSR server infrastructure is owned and hosted entirely by HP. The first-time use of the HP Virtual Support Room will initiate a small client download (less than 2Mb).

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Outbound Security

All HP RDA Solutions are designed to be used for inbound access from HP to customer networks. All RDA solutions, with the exception of the Virtual CAS, do not initiate outbound connections without direct user interaction. Confidentiality for outbound connections is provided by the connection service (SSL over HTTPS, SSH, IPsec etc). Authentication mechanisms can vary from solution to solution, but all solutions are designed to ensure the privacy and security of all parties. The Virtual Customer Access System (vCAS) initiates outbound connections to VeriSign.com to validate certificates, using either OCSP to check the CRL status of an individual certificate, or HTTP to periodically fetch the entire CRL for the HP Class 2 Certification Authority. The Virtual CAS also periodically connects to the HP repository server using HTTPS to check for and fetch software updates.

Inbound Security

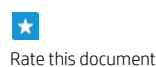
Remote Device Access requires an inbound connection from HP to a customer-designated access server. HP understands that IT security policies within organizations vary considerably. Therefore, HP offers a number of remote access solutions (depending on the service level agreement) designed to meet customer security requirements. All of HP solutions use standard techniques that include SSH, IPsec, and HTTPS. HP offers both hardware and software solutions which can be configured to ensure that the customer is always in control of the connection. HP also has options that allow the customer to view and monitor a support specialist's activities.

All HP support specialists must adhere to the same standards of business conduct as onsite HP engineers, and are only allowed to initiate a connection with the customer's approval and a valid business need. Access restrictions can be placed on specific connection profiles to limit HP's access to a subset of support personnel. Access restrictions can be restricted by region and/or country. It can also be restricted to HP support personnel for a specific product platform. Access controls can also be restricted to specific HP personnel. Access controls can be enforced both at HP (before the connection is initiated) and again at the CAS (see the vCAS solution). This model ensures that both the HP Account Manager and the customer administrator can control HP access to the customer network. Internally, HP uses two-factor authentication to control access through the HP Remote Access Connectivity (RACS). Additionally, all connections, attempted and successful, to customer systems are logged.

Security Auditing

All attended RDA connection attempts from HP to customers are logged. The acting user, start and stop times of the connection, and the connection status are logged. The connection status will indicate failures such as improper authentication and authorization. This tracking information is retained for 13 months.

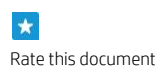
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GLOSSARY of Terms

API	Application Programming Interface
DCOM	Distributed Component Object Module
EDW	Enterprise Data Warehouse
ELMC	Event Log Monitoring Collector
ESP	Encapsulating Security Payload
GDID	Global Support Identifier
GUI	Graphical User Interface (same as UI)
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol Secure
IKEv2	Internet Key Exchange version 2
IP	Internet Protocol
IPSEC	Internet Protocol Security
LAN	Local Area Network
OSCP	Online Certificate Status Protocol
RDA	Remote Device Access
RDC	Remote Data Collection
RDM	Remote Device Management
RIBCL	Remote Insight Board Control Language
RS	Remote Support
RSDB	Remote Support Database
SADB	Support Automation Database
SLA	Service Level Agreement
SNMPv1	Simple Network Management Protocol Version 1
SNMPv2	Simple Network Management Protocol Version 2
SNMPv3	Simple Network Management Protocol Version 3
SSL	Secure Sockets Layer
TCP	Transmission Control Protocol
TLS	Transport Layer Security
UDP	Unified Datagram Protocol
UI	User Interface
WEBM	Web-Based Enterprise Management
WMI	Windows Measurement Instrumentation

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Appendix A: Summary of Network Ports for Standard Operating System Connectivity

The following tables summarize all ports that might be used in Insight Remote Support Hosting Device Operating System Connectivity. The following ports are required for basic system operation.

A.1 Standard Operating System Network Ports

Table A.1 Standard Operating System Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
UDP	53	System	DNS Server	Domain Name Service (DNS) - Host name resolution.	No	Required
UDP	123	System	NTP Server	Network Time Protocol - Synchronizes system clock	No	Recommended
TCP	3389	Customer's MSRD (Terminal Services) Client	System	Remote Desktop Protocol - Remote management (to change port number, see http://support.microsoft.com/kb/306759)	Yes	Optional
TCP	22	Customer's SSH Client	System	Secure Shell - Remote management	Yes	Optional
TCP	80 or web proxy port	Customer's Web Browser	Web Server or Web Proxy	HTTP web access	Yes	Optional
TCP	443 or web proxy port	Customer's Web Browser	Web Server or Web Proxy	HTTPS web access	Yes	Optional
TCP	25	System	SMTP Server	Simple Mail Transfer Protocol - Sending email	No	Optional

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Appendix B: Summary of Network Ports for Servers

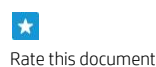
The following tables summarize all ports that might be used in Insight Remote Support for Servers. See Table A-1 for ports that are required for basic system operation.

B.1 Hosting Device

Table B.1 Hosting Device Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	25	Hosting Device	Customer-Designated SMTP Server	Email notifications	No	Recommended
TCP	443 or web proxy port	Hosting Device	services.isee.hp.com or Web Proxy	Transport of hardware events and data collections to HP, synchronization of submission status from HP to Hosting Device, requests for device contract, warranty information and health-check data	No	Required
TCP	2381	Customer's Web Browser	Hosting Device	HP SMH secure web server (HTTPS) and RDC from monitored systems	No	Recommended
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required
TCP	80 or web proxy port	Hosting Device	software.hp.com or Web Proxy	Software application download	No	Recommended
TCP	3389	Customer's MSRDP (Terminal Services) Client	Hosting Device	Microsoft Remote Desktop Connection (RDC) used for remote management by HP or customer	No	Optional

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B.2 HP-UX Monitored Systems

Table B.2 HP-UX Connectivity - Firewall/Port Requirements

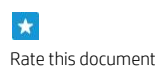
Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required
TCP	2381	Hosting Device	Monitored Systems	HP SMH secure web server (HTTPS) and RDC from monitored systems	No	Optional

B.3 Integrity Linux Monitored Systems

Table B.3 Integrity Linux Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

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B.4 Integrity Windows Server 2003 Monitored Systems

Table B.4 Integrity Windows Server 2003 Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Required
TCP	7920	Hosting Device	Monitored Systems	The Insight-RS ELMC (formerly WCCProxy) process communicates with the Director on this port. This is a proprietary protocol. Any connections that exchange username and passwords use SSL. Not all connections are SSL.	No	Required
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	No	Required
TCP	135	Monitored Systems	Hosting Device	DCE endpoint resolution. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
TCP	139	Monitored Systems	Hosting Device	NETBIOS Session Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
TCP	1024-65535	Monitored Systems	Hosting Device	Windows Server 2003 Windows Management Interface (WMI) Communications DCOM dynamic port assignment. Note that the Hosting Device can be configured to limit this range. The source port will always be 135.	No	Required
UDP	137	Monitored Systems	Hosting Device	NETBIOS Name Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
UDP	138	Monitored Systems	Hosting Device	NETBIOS Datagram Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
UDP	445	Monitored Systems	Hosting Device	Microsoft File Sharing. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required

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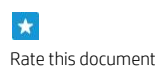
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required
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B.5 Integrity Windows Server 2008 Monitored Systems

Table B.5 Integrity Windows Server 2008 Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Required
TCP	135	Monitored Systems	Hosting Device	DCE endpoint resolution. Used by DCOM, and hence, Windows Management Interface (WMI) and INSIGHT-RS	No	Required
TCP	139	Monitored Systems	Hosting Device	NETBIOS Session Service. Used by DCOM, and hence, Windows Management Interface (WMI) and INSIGHT-RS	No	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
TCP	49152 - 65535	Monitored Systems	Hosting Device	Windows Server 2008 Windows Management Interface (WMI) Communications DCOM dynamic port assignment. Note that the Hosting Device can be configured to limit this range. The source port will always be 135.	No	Required
UDP	137	Monitored Systems	Hosting Device	NETBIOS Name Service. Used by DCOM, and hence, Windows Management Interface (WMI) and INSIGHT-RS	No	Required
UDP	138	Monitored Systems	Hosting Device	NETBIOS Datagram Service. Used by DCOM, and hence, Windows Management Interface (WMI) and INSIGHT-RS	No	Required
UDP	445	Monitored Systems	Hosting Device	Microsoft File Sharing. Used by DCOM, and hence, Windows Management Interface (WMI) and INSIGHT-RS	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

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B.6 OpenVMS Integrity Monitored Systems

Table B.6 OpenVMS Integrity Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Required
TCP	7920	Hosting Device	Monitored Systems	The Insight-RS ELMC (formerly WCCProxy) process communicates with the Director on this port. This is a proprietary protocol. Any connections that exchange username and passwords use SSL. Not all connections are SSL.	No	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

B.7 ProLiant C-Class Blade Enclosure

Table B.7 ProLiant C-Class Blade Enclosure Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	443	Hosting Device	Onboard Administrator	RIBCL Event Listener	No	Required
UDP	161	Hosting Device	Onboard Administrator	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
UDP	162	Onboard Administrator	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
TCP	7906	Onboard Administrator	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7906)	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

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B.8 ProLiant Citrix Monitored Systems

Table B.8 ProLiant Citrix Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
TCP	443	Hosting Device	Onboard Administrator/iLO4	RIBCL Event Listener	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

B.9 ProLiant Generation 8 Monitored Systems

ProLiant Generation 8 Connectivity - Firewall/Port Requirements*

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	443	Hosting Device	Onboard Administrator/iLO4	RIBCL Event Listener	No	Required
TCP	7906	Onboard Administrator/iLO4	Hosting Device	Secure HTTP (HTTPS) port used by the listener running on the Embedded Support Device. The management device connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7906)	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

* Hardware requirements only, Operating System port requirements should also be configured

B.10 ProLiant Linux Monitored Systems

ProLiant Linux Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

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B.11 ProLiant Microsoft Hyper-V Monitored Systems

ProLiant Microsoft Hyper-V Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Required
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
TCP	135	Monitored Systems	Hosting Device	DCE endpoint resolution. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
TCP	139	Monitored Systems	Hosting Device	NETBIOS Session Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
TCP	49152-65535	Monitored Systems	Hosting Device	Windows Server 2008 Windows Management Interface (WMI) Communications DCOM dynamic port assignment. Note that the Hosting Device can be configured to limit this range. The source port will always be 135.	No	Required
UDP	137	Monitored Systems	Hosting Device	NETBIOS Name Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
UDP	138	Monitored Systems	Hosting Device	NETBIOS Datagram Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
UDP	445	Monitored Systems	Hosting Device	Microsoft File Sharing. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

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B.12 ProLiant VMWare ESX Monitored Systems

ProLiant VMWare ESX Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

B.13 ProLiant VMWare ESXi Monitored Systems

ProLiant VMWare ESXi Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

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B.14 ProLiant Windows Server 2003 Monitored Systems

ProLiant Windows Server 2003 Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Required
TCP	135	Monitored Systems	Hosting Device	DCE endpoint resolution. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
TCP	139	Monitored Systems	Hosting Device	NETBIOS Session Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
TCP	1024-65535	Monitored Systems	Hosting Device	Windows Server 2003 Windows Management Interface (WMI) Communications DCOM dynamic port assignment. Note that the Hosting Device can be configured to limit this range. The source port will always be 135.	No	Required
TCP	49152-65535	Monitored Systems	Hosting Device	Windows Server 2008 Windows Management Interface (WMI) Communications DCOM dynamic port assignment. Note that the Hosting Device can be configured to limit this range. The source port will always be 135.	No	Required
UDP	137	Monitored Systems	Hosting Device	NETBIOS Name Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
UDP	138	Monitored Systems	Hosting Device	NETBIOS Datagram Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
UDP	445	Monitored Systems	Hosting Device	Microsoft File Sharing. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Optional
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Optional

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B.15 ProLiant Windows Server 2008 Monitored Systems

ProLiant Windows Server 2008 Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Required
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
TCP	135	Monitored Systems	Hosting Device	DCE endpoint resolution. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
TCP	139	Monitored Systems	Hosting Device	NETBIOS Session Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
TCP	49152-65535	Monitored Systems	Hosting Device	Windows Server 2008 Windows Management Interface (WMI) Communications DCOM dynamic port assignment. Note that the Hosting Device can be configured to limit this range. The source port will always be 135.	No	Required
UDP	137	Monitored Systems	Hosting Device	NETBIOS Name Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
UDP	138	Monitored Systems	Hosting Device	NETBIOS Datagram Service. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
UDP	445	Monitored Systems	Hosting Device	Microsoft File Sharing. Used by DCOM, and hence, Windows Management Interface (WMI) and Insight-RS	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

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Appendix C: Summary of Network Ports for Storage

The following tables summarize all ports that might be used in Insight Remote Support for Storage. See Table A-1 for ports that are required for basic system operation.

C.1 StorageWorks MSA15XX/2XXX G1 Storage Systems

Table C.1 StorageWorks MSA15XX/2XXX G1 Storage Systems Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	2301	Customer's Web Browser	Hosting Device	HP SMH port for Insight Manager Web Agents; HTTP (unencrypted) ? redirected to 2381 (HTTPS)	Yes	Required
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

C.2 StorageWorks MSA23xx G2 Storage Systems

Table C.2 StorageWorks MSA23xx G2 Storage Systems Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	2301	Customer's Web Browser	Hosting Device	HP SMH port for Insight Manager Web Agents; HTTP (unencrypted) ? redirected to 2381 (HTTPS)	Yes	Required
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Optional

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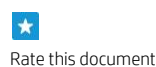


C.3 HP P4000 Storage Systems

Table C.3 HP P4000 Storage Systems Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	5989	CMC (can be running on Hosting Device)	Monitored Systems	HP P4000 Centralized Management Console (CMC)	No	Required
TCP	5989	Hosting Device	Monitored Systems	Remote Support P4000 Integration Module - HP P4000 CLI API	Yes	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

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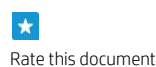


D.4 StorageWorks P6000 (EVA) Storage Systems

Table D.4 EVA Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	2372	Hosting Device	EVA	P6000/EVA CommandView - Storage Collections for EVA (HTTPS)	No	Required
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used to communicate with WBEM end point nodes.	Yes	Required
TCP	7920	Hosting Device	Monitored Systems	The Insight-RS ELMC (formerly WCCProxy) process communicates with the Director on this port. This is a proprietary protocol. Any connections that exchange username and passwords use SSL. Not all connections are SSL.	No	Required
TCP	2373	Hosting Device	P6000 ABM	P6000/EVA CommandView - Storage Collections for P6000/EVA Array Based Management	No	Required
TCP	7905	Monitored Systems	Hosting Device	Secure HTTP (HTTPS) port used by the listener running in the Director's Web Interface. The monitored host connects to the Hosting Device on this port (e.g. https://target.sys.name.here:7905)	No	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Optional
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Optional

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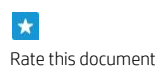


C.5 StorageWorks Tape Libraries

Table C.5 StorageWorks Tape Libraries Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	2301	Customer's Web Browser	Hosting Device	HP SMH port for Insight Manager Web Agents; HTTP (Recommend using TCP/2381)	Yes	Optional
TCP	2381	Customer's Web Browser	Hosting Device	HP SMH port for Insight Manager Web Agents; HTTPS redirected to 2381 (HTTPS)	Yes	Recommended
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
TCP	5989	Hosting Device	Monitored Systems	Secured WBEM CI-MOM protocol over HTTPS/SOAP. This port is used for basic collections on HP StoreEver ESL & EML Devices	Yes	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

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Appendix E: Summary of Network Ports for Networking

The following tables summarize all ports that might be used in Insight Remote Support for Networking. See Table A-1 for ports that are required for basic system operation.

E. 1 A-Series/E-Series Switch Monitored Systems

Table E. 1 A-Series/E-Series Switch Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required
TCP	22	Hosting Device	Monitored Systems	SSH: Remote Data Collection	Yes	Optional

E.2 SAN Monitored Systems

Table E.2 SAN Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required
TCP	22	Hosting Device	Monitored Systems	SSH: SAN Data Collection (HP VLS, HP StoreEver ESL & MSL Devices)	No	Recommended
TCP	23	Hosting Device	Monitored Systems	Telnet (unencrypted): SAN Data Collection (HP StoreOnce D2D Backup & optionally by HP StoreEver EML/ESL Devices)	No	Optional
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port. Used by HP StoreEver MSL, HP C-Series and H-Series SAN Switches for San Data Collections	Yes	Optional
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Optional

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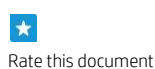


E.3 SAN Switch Monitored Systems

Table E.3 SAN Switch Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required
ICMP	N/A	Hosting Device	Monitored Systems	Provides system reachability (ping) check during system discovery and before other operations.	No	Required

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Appendix F: Summary of Network Ports for HP UPS Management Module Connectivity

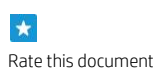
The following tables summarize all ports that might be used in Insight Remote Support for HP UPS Management Module Connectivity. See Table A-1 for ports that are required for basic system operation.

F.1 HP UPS Management Module Connectivity

Table G.1 HP UPS Management Module Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required

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Appendix G: Summary of Network Ports for Remote Device Access

The following tables summarize all ports that might be used in Remote Device Access. See Table A-1 for ports that are required for basic system operation.

G.1 Customer Access System (CAS)

Table H.1 CAS Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	22	HP Remote Access Connectivity System (RACS)	CAS	SSH Tunnel (SSH-Direct only)	No	Required for SSH-Direct
ICMP	N/A	CAS	Target System Including Hosting Device	Provides system reachability (ping) check during installation	No	Recommended
ICMP	N/A	Target System Including Hosting Device	CAS	Provides system reachability (ping) check during installation.	No	Recommended
TCP	22	CAS	Target System Including Hosting Device	SSH command-line access	Yes	Optional
TCP	23	CAS	Target System Including Hosting Device	Telnet command-line access if SSH is not available.	Yes	Optional
TCP	80	CAS	Target System Including Hosting Device	HTTP connection forwarded from HP through CAS to Hosting Device or monitored system	Yes	Optional
TCP	3389	CAS	Target System Including Hosting Device	MS RDP. Remote Desktop Connection forwarded from HP through CAS to Hosting Device or monitored system	Yes	Optional
TCP	5800	CAS	Target System Including Hosting Device	VNC Web access	Yes	Optional
TCP	5900	CAS	Target System Including Hosting Device	VNC access	Yes	Optional
TCP	other	CAS	Target System Including Hosting Device	Customer-specified port and application protocol SSH-forwarded from HP	Yes	Optional
TCP	other	Customer Clients	CAS	Other access methods for CAS administration	Yes	Optional

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TCP	22	Customer's SSH Client	Target System Including Hosting Device	SSH Command-line access	Yes	Optional
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G.2 Additional Ports for Virtual CAS

Table H.2 Additional Ports for Virtual CAS Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	443	Customer's Web Browser	Virtual CAS	HTTPS port for web UI for managing Virtual CAS	No	Required
UDP	53	Virtual CAS	DNS Server	Domain Name Service (DNS) - Host name resolution	No	Required
UDP	123	Virtual CAS	Network Time Server	Network Time Protocol	No	Recommended
TCP	80 or web proxy port	Virtual CAS	onsitecrl.verisign.com or Web Proxy	HTTP (Unencrypted) Daily fetch of HP Class 2 CA certificate revocation list (CRL)	No	Recommended
TCP	80	Virtual CAS	onsite-ocsp.verisign.com	OCSP (Online Certificate Status Protocol) for certificate revocation check	No	Recommended
TCP	22	Customer's SSH Client	Virtual CAS	SSH command-line access for Virtual CAS management	No	Optional
TCP	25	Virtual CAS	Customer-Designated SMTP Server	Email notifications	No	Optional
TCP	443 or web proxy port	Virtual CAS	h20529.www2.hp.com or Web Proxy	HTTPS connection to the HP RDA CAS Kit server to download updates	No	Optional
TCP	514	Virtual CAS	Logging Server	Syslog remote logging (unencrypted)	Yes	Optional
UDP	514	Virtual CAS	Logging Server	Syslog remote logging (unencrypted)	Yes	Optional
TCP	other	Virtual CAS	Target System	Customer-specified TCP port and application protocol SSH-forwarded from HP via the relay application	Yes	Optional
UDP	other	Virtual CAS	Target System	Customer-specified UDP port and application protocol SSH-forwarded from HP via the relay application	Yes	Optional

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G.3 Additional Ports for iCAS

Table H.3 Additional Ports for iCAS Connectivity - Firewall/Port Requirements

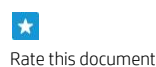
Protocol	Ports	Source	Destination	Function	Configurable	Optional
UDP	53	iCAS Host	DNS Server	Domain Name Service (DNS) - Host name resolution	No	Required
TCP	80 or web proxy port	iCAS Host	HP Regional RAMS Server or Web Proxy	HTTP Tunnelling for SSH (tcp/2022)	No	Optional
TCP	443 or web proxy port	iCAS Host	HP Regional RAMS Server or Web Proxy	HTTPS to retrieve iCAS plug-in and HTTPS Tunnelling for SSH (tcp/2022)	No	Recommended
TCP	other	iCAS Host	Target System	Customer-specified TCP port and application protocol SSH-forwarded from HP	Yes	Optional
UDP	other	iCAS Host	Target System	Customer-specified UDP port and application protocol SSH-forwarded from HP	Yes	Optional

G.4 Additional Ports for P9000/XP Storage Array

Additional Ports for P9000/XP Storage Array Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
TCP	3389	CAS	XP SVP	Microsoft Remote Desktop Connection (RDC) used for remote management by HP or customer	No	Recommended
TCP	5631	CAS	XP SVP	pcAnywhere data connection for accessing XP SVP. In case of XP1024/XP128, Pcan anywhere is the only option to access SVP since RDC is not supported for Win2k. Engineers may have to first login to XPSVP using pcan anywhere to know which Hosting Device the XP is pointed.	Yes	Optional
UDP	5632	CAS	XP SVP	pcAnywhere status for accessing XP SVP	Yes	Optional

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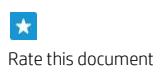
Appendix H: Summary of Network Ports for HP UPS Management Module Connectivity

H.1 HP UPS Management Module Connectivity

HP UPS Management Module Connectivity - Firewall/Port Requirements

Protocol	Ports	Source	Destination	Function	Configurable	Optional
UDP	161	Hosting Device	Monitored Systems	SNMP. This is the standard port used by SNMP agents on monitored systems. The Hosting Device sends requests to devices on this port.	Yes	Required
UDP	162	Monitored Systems	Hosting Device	SNMP Trap. This is the standard port used by SNMP managers for listening to traps.	Yes	Required

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[RFC 4302: IP Authentication Header](#)

[RFC 4303: IP Encapsulating Security Payload \(ESP\)](#)

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[RFC 6277: Online Certificate Status Protocol Algorithm Agility](#)

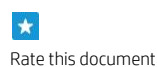
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